Nonbranch weighted-Proj stalk: checked local boundary

Formal proof packet

28 October 2026

This report records a checked theorem and its explicit hypotheses. It adds no experiment, target result, or cost ratio. The editable SVG, standardized CSV, Lean source, and compiler log accompany this PDF.

Checked statements

For any field K, quintic $f(x) = x^5 + a_3x^3 + a_2x^2 + a_1x + a_0$, and rational pair (a, y) with $y^2 = f(a)$, the Lean file constructs the point on the actual weighted Proj curve through its Z basic open. The theorem `pointProjStalkIso` identifies its structure-sheaf stalk with the affine coordinate-ring localization at the evaluation kernel. If $y \neq 0$, `pointY_unit` proves the pulled-back affine y is a unit in that stalk.

The theorem `point_norm_order` is checked on this actual stalk *with explicit* `IsDomain` and `IsDiscreteValuationRing` instances, and with 2, B, and y units, $N - By$ a nonunit, and $(N - By)(N + By) = u_1u_2U$. It concludes

$$\operatorname{ord}(N - By) = \operatorname{ord}(u_1) + \operatorname{ord}(u_2) + \operatorname{ord}(U).$$

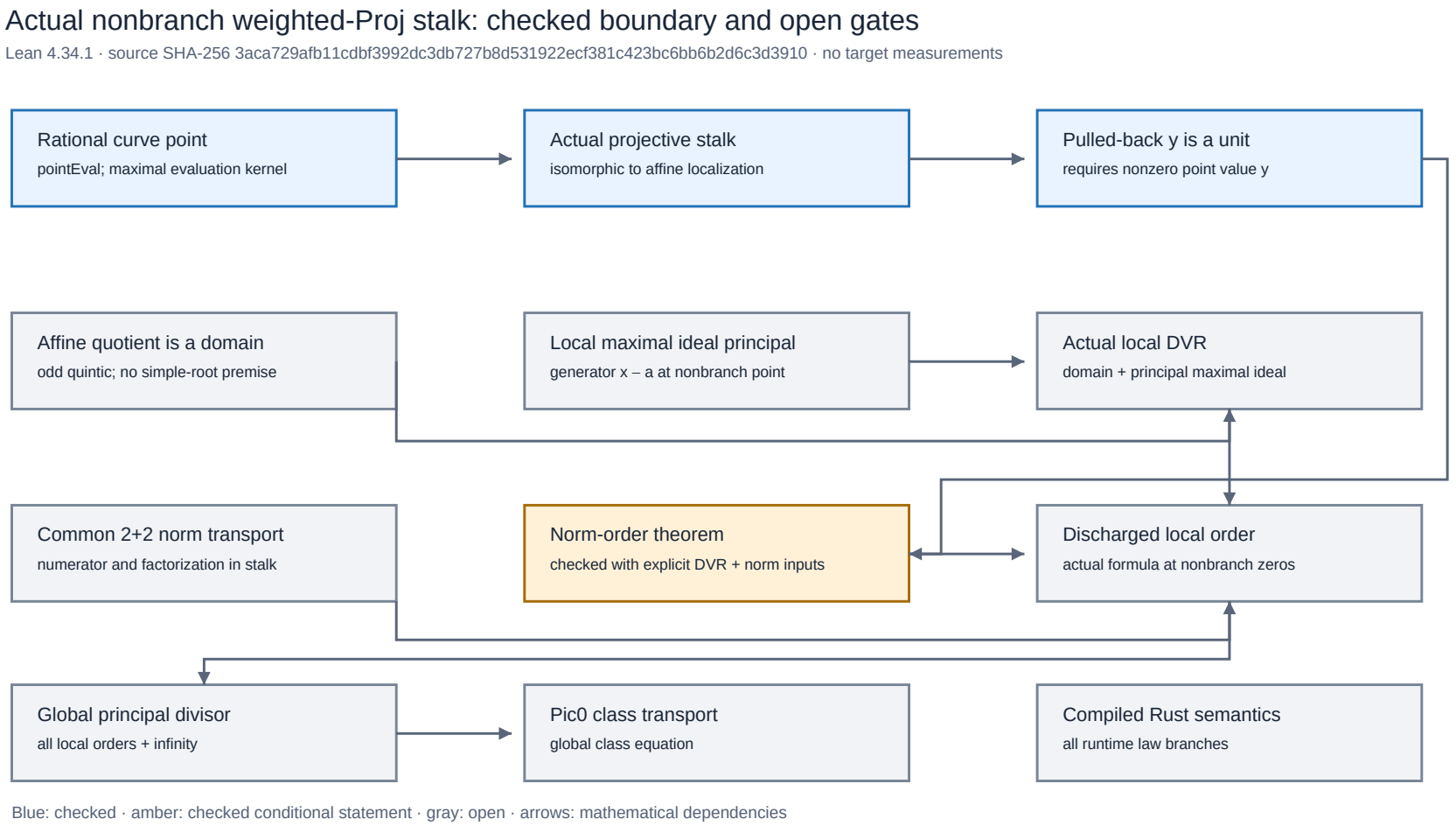
This applies the previously checked abstract local norm-order lemma; it does not discharge the DVR or formula-specific norm premises.

Verification and open gates

Direct compilation with pinned Lean 4.34.1 exited zero. The pinned Mathlib revision is `d13f23b723b8a846827a245b89c10fc7d3f11612`. Seven printed theorem audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Lean source SHA-256 is `3aca729afb11cdbf3992dc3db727b8d531922ecf381c423bc6bb6b2d6c3d3910`.

The affine odd-quintic domain proof, principal nonbranch maximal ideal, and actual DVR instance remain open. So do transport of the proof-side common 2+2 numerator and norm factorization, a global principal divisor, Pic^0 class transport, and compiled Rust semantics. The exact row-by-row status is in `nonbranch-stalk-order.csv`. Stalk transport follows the open-immersion principle in Stacks Project Tag 01HD; the proposed local DVR criterion is Tag 00PD.

Proof dependencies and gates



Blue: checked; amber: checked conditional statement; gray: open. Arrows are proof dependencies. The diagram contains no performance data.